

APEXPLANET CYBERSECURITY INTERNSHIP

TASK 5

DVWA Web Application Penetration Testing
and Incident Response

Capstone Project Report

Controlled Local Laboratory Environment

Prepared from the submitted Task-5 project files and evidence screenshots.

1. Executive Summary

This report documents a controlled security assessment of the Damn Vulnerable Web Application (DVWA) hosted in a local Kali Linux laboratory. The project covers planning, service reconnaissance, web application assessment, evidence collection, SQL Injection validation, Apache log analysis, simulated containment and recovery, findings, mitigations, and post-incident documentation.

Assessment boundary: The submitted project states that testing was restricted to the authorized local/host-only laboratory environment and did not target production or external systems.

2. Project Objectives and Scope

APEXPLANET CYBERSECURITY INTERNSHIP - TASK 5

Project Title:

Web Application Penetration Testing & Incident Response on DVWA

Objective:

Assess a deliberately vulnerable DVWA web application in a controlled lab environment, identify security weaknesses, document evidence, recommend mitigations, and simulate an incident response workflow.

Scope:

Target: DVWA

Environment: Kali Linux + DVWA

Network: Private/Host-Only VirtualBox Lab

Testing is limited to the authorized lab environment.

Tools:

- Kali Linux
- DVWA
- Burp Suite
- Nmap
- Browser Developer Tools
- curl

Planned Assessment Areas:

1. Reconnaissance
2. Application and service discovery
3. SQL Injection assessment
4. Cross-Site Scripting assessment
5. CSRF assessment
6. File Inclusion assessment where applicable
7. Security headers review
8. Evidence collection
9. Mitigation recommendations
10. Incident response simulation

Incident Response:

- Detect simulated attack activity
- Analyze available logs/evidence
- Contain the simulated threat
- Recommend eradication/remediation
- Prepare a post-incident report

Final Deliverables:

- Capstone Project Report PDF

- GitHub Repository
- 12-minute Final Presentation Video

3. Lab Architecture / Network Diagram

APEXPLANET TASK 5

DVWA WEB APPLICATION PENTEST LAB

■ Kali Linux ■

■ Pentest Workstation ■

■ 192.168.56.101 ■

■

- Host-Only /

- Local Lab

▼

■ Apache ■

■ Web Server ■

■ HTTP : 80 ■

■

▼

■ DVWA ■

■ Damn Vulnerable Web ■

■ Application ■

▼

■ MariaDB ■

■ Database: dvwa ■

Scope: Controlled local lab environment

Purpose: Authorized web application security testing

The architecture describes Kali Linux as the pentest workstation, Apache as the web server, DVWA as the intentionally vulnerable application, and MariaDB as its database in a controlled lab.

4. Methodology

- Project planning and scope definition
- Lab and application/service verification
- Network and service reconnaissance
- HTTP/web-server assessment
- Controlled SQL Injection validation
- Apache access-log analysis and attack-indicator identification
- Simulated containment and recovery
- Findings, mitigation and post-incident documentation

5. Technical Evidence Summary

HTTP headers:

```
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 17:10:30 GMT
Server: Apache/2.4.68 (Debian)
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referrer-Policy: strict-origin-when-cross-origin
Permissions-Policy: geolocation=(), microphone=(), camera=()
Set-Cookie: security=impossible; path=/; HttpOnly
Set-Cookie: PHPSESSID=ee28e81981a8d0267a414823b8c79f40; expires=Wed, 16 Sep 2026 17:10:30 GMT;
Max-Age=86400; path=/; HttpOnly; SameSite=Strict
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Set-Cookie: PHPSESSID=77c2c82df986d65cbb9523f175b34b0c; expires=Wed, 16 Sep 2026 17:10:30 GMT;
Max-Age=86400; path=/; HttpOnly; SameSite=Strict
Location: login.php
Content-Type: text/html; charset=UTF-8
```

Apache access-log evidence:

```
127.0.0.1 - - [15/Sep/2026:12:50:03 -0400] "GET /DVWA/index.php HTTP/1.1" 200 2974
"http://127.0.0.1/DVWA/login.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101
Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:50:10 -0400] "GET /DVWA/ HTTP/1.1" 200 2974 "-" "Mozilla/5.0 (X11; Linux
x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:03 -0400] "GET /DVWA/security.php HTTP/1.1" 200 2431
"http://127.0.0.1/DVWA/index.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101
Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:04 -0400] "GET /DVWA/dvwa/images/lock.png HTTP/1.1" 304 422
"http://127.0.0.1/DVWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101
Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:18 -0400] "POST /DVWA/security.php HTTP/1.1" 302 673
"http://127.0.0.1/DVWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101
Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:18 -0400] "GET /DVWA/security.php HTTP/1.1" 200 2443
"http://127.0.0.1/DVWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101
Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:55:46 -0400] "HEAD /DVWA/ HTTP/1.1" 302 777 "-" "curl/8.20.0"
127.0.0.1 - - [15/Sep/2026:12:57:52 -0400] "GET /DVWA/vulnerabilities/sqli/ HTTP/1.1" 200 1963
"http://127.0.0.1/DVWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101
Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:58:44 -0400] "GET /DVWA/vulnerabilities/sqli/?id=1&Submit=Submit HTTP/1.1"
200 1986 "http://127.0.0.1/DVWA/vulnerabilities/sqli/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0)
Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:13:01:04 -0400] "GET
/DVWA/vulnerabilities/sqli/?id=1%27+OR+%271%27%3D%271&Submit=Submit HTTP/1.1" 200 2044
"http://127.0.0.1/DVWA/vulnerabilities/sqli/?id=1&Submit=Submit" "Mozilla/5.0 (X11; Linux x86_64;
rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:13:08:10 -0400] "GET / HTTP/1.0" 200 11151 "-" "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "POST /sdk HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible;
Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /nmaplowercheck1789492101 HTTP/1.1" 404 665 "-"
"Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-" "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /HNAP1 HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible;
Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /evox/about HTTP/1.1" 404 665 "-" "Mozilla/5.0
(compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-" "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.1" 200 11132 "-" "-"
127.0.0.1 - - [15/Sep/2026:13:10:30 -0400] "HEAD /DVWA/ HTTP/1.1" 302 777 "-" "curl/8.20.0"
```

6. Findings and Mitigations

APEXPLANET TASK 5 — SECURITY FINDINGS & MITIGATIONS

Finding 1: SQL Injection

Severity: High

Evidence: SQL Injection request was recorded in the Apache access log.

Risk: Improperly handled user input may allow manipulation of database queries.

Mitigation: Use prepared/parameterized SQL statements, validate input, and apply least-privilege database access.

Finding 2: Web Application Exposure

Severity: Medium

Evidence: DVWA was accessible through the local Apache web server.

Risk: A deliberately vulnerable application should not be exposed outside the controlled testing environment.

Mitigation: Restrict access to authorized networks, use firewall rules, and never deploy DVWA in a production environment.

Finding 3: Security Configuration

Severity: Informational

Evidence: HTTP response headers were reviewed during the assessment.

Observed protections included X-Content-Type-Options, X-Frame-Options, Referrer-Policy and Permissions-Policy.

Mitigation: Maintain secure headers and periodically review the complete web-server security configuration.

Finding 4: Incident Detection

Severity: Medium

Evidence: Apache access logs contained a suspicious SQL Injection request.

Risk: Without log monitoring, malicious requests may not be detected promptly.

Mitigation: Centralize logs, create detection rules for suspicious patterns, and establish an incident-response process.

Overall Assessment:

The assessment demonstrated security weaknesses and detection opportunities within the controlled DVWA laboratory. The findings are limited to the local training environment and do not represent a production system assessment.

7. Incident Response

APEXPLANET TASK 5 — POST-INCIDENT REPORT

Incident Type:

Simulated Web Application SQL Injection

Environment:

Local DVWA lab on Kali Linux

Detection:

Apache access logs were reviewed and a suspicious SQL Injection request pattern was identified in the DVWA SQL Injection endpoint.

Indicator:

Encoded SQL metacharacters and OR-based SQL expression were present in the HTTP request.

Containment:

Apache HTTP Server was temporarily stopped to isolate the affected local web application during the simulated incident-response exercise.

Eradication:

The simulated threat was treated as a controlled lab event.
No malicious persistence or malware was introduced into the system.

Recovery:

Apache HTTP Server was restarted and verified as active and running.

Impact:

The activity was restricted to the local DVWA training environment.
No production or external system was involved.

Recommended Mitigations:

- Use parameterized/prepared SQL statements.
- Validate and sanitize user input.
- Apply least-privilege database permissions.
- Monitor web-server logs for suspicious request patterns.
- Maintain secure application configuration.
- Perform regular vulnerability assessments.

Lessons Learned:

Log monitoring can help identify suspicious application requests.
Containment limits exposure while an incident is investigated.
Post-incident documentation helps track detection, response, recovery, and future preventive actions.

Attack Timeline

APEXPLANET TASK 5 — ATTACK TIMELINE

1. Lab Preparation

- DVWA and supporting services verified.
- Apache web server and MariaDB database confirmed.

2. Reconnaissance

- Local DVWA environment identified.
- HTTP service and database service assessed.

3. Web Assessment

- HTTP response headers reviewed.
- DVWA SQL Injection functionality assessed.

4. Detection

- Apache access logs reviewed.
- Suspicious SQL Injection request identified.

5. Containment

- Apache service temporarily stopped as a simulated containment action.

6. Recovery

- Apache service restarted.
- Service status verified as active and running.

7. Post-Incident Analysis

- Findings documented.
- Security improvements and mitigation recommendations recorded.

8. Remediation Plan

APEXPLANET TASK 5 — REMEDIATION PLAN

1. SQL Injection

- Use prepared statements / parameterized queries.
- Validate and sanitize all user-supplied input.
- Avoid dynamically constructing SQL queries.
- Apply least-privilege permissions to the application database account.

2. Web Application Security

- Keep vulnerable training applications isolated from production networks.
- Restrict access using firewall and network controls.
- Keep Apache, PHP and application components updated.

3. Security Headers

- Maintain secure HTTP response headers.
- Periodically review headers and remove unnecessary information exposed by the web server.

4. Logging and Monitoring

- Monitor Apache access and error logs.
- Detect suspicious encoded characters and abnormal request patterns.
- Retain logs for incident investigation.

5. Incident Response

- Define detection, containment, eradication and recovery procedures.
- Isolate affected services during investigation.
- Restore services only after the environment has been checked.

6. General Security

- Perform regular vulnerability assessments.
- Follow secure coding practices.
- Document findings, evidence and remediation actions.

9. Repository Documentation

ApexPlanet Task 5 — DVWA Web Application Pentest

Project Overview

This project presents a controlled web application security assessment of the Damn Vulnerable Web Application (DVWA) in a local Kali Linux lab.

Objective

The objective was to assess the security of the DVWA application, identify security weaknesses, collect technical evidence, and document appropriate mitigation and incident-response actions.

Scope

- Target: Local DVWA installation
- Environment: Kali Linux
- Web Server: Apache
- Database: MariaDB
- Testing restricted to the authorized local laboratory environment

Tools Used

- Kali Linux
- DVWA
- Apache
- MariaDB
- Nmap
- cURL
- Browser
- Apache access logs

Methodology

1. Project planning
2. Lab and application verification
3. Network and service reconnaissance
4. Web application assessment
5. Evidence collection
6. Finding and risk documentation
7. Incident detection through access logs
8. Simulated containment and recovery
9. Post-incident documentation

Key Findings

SQL Injection

A controlled SQL Injection test was performed against the DVWA SQL Injection functionality. The corresponding request was also identified in the Apache access log.

Web Application Exposure

The intentionally vulnerable DVWA application was accessible through the local Apache web server.

Log Monitoring

Apache access logs provided useful evidence for identifying suspicious requests during the assessment.

Incident Response

The simulated SQL Injection activity was identified through Apache logs. Apache was temporarily stopped as a containment action and subsequently restarted to recover the lab environment.

Mitigation

- Use prepared/parameterized SQL statements.
- Validate and sanitize user input.
- Apply least-privilege database permissions.
- Monitor web-server logs for suspicious requests.
- Restrict vulnerable applications to authorized testing environments.
- Maintain secure web-server configuration.

Evidence

Screenshots and supporting files are maintained in the project evidence directory.

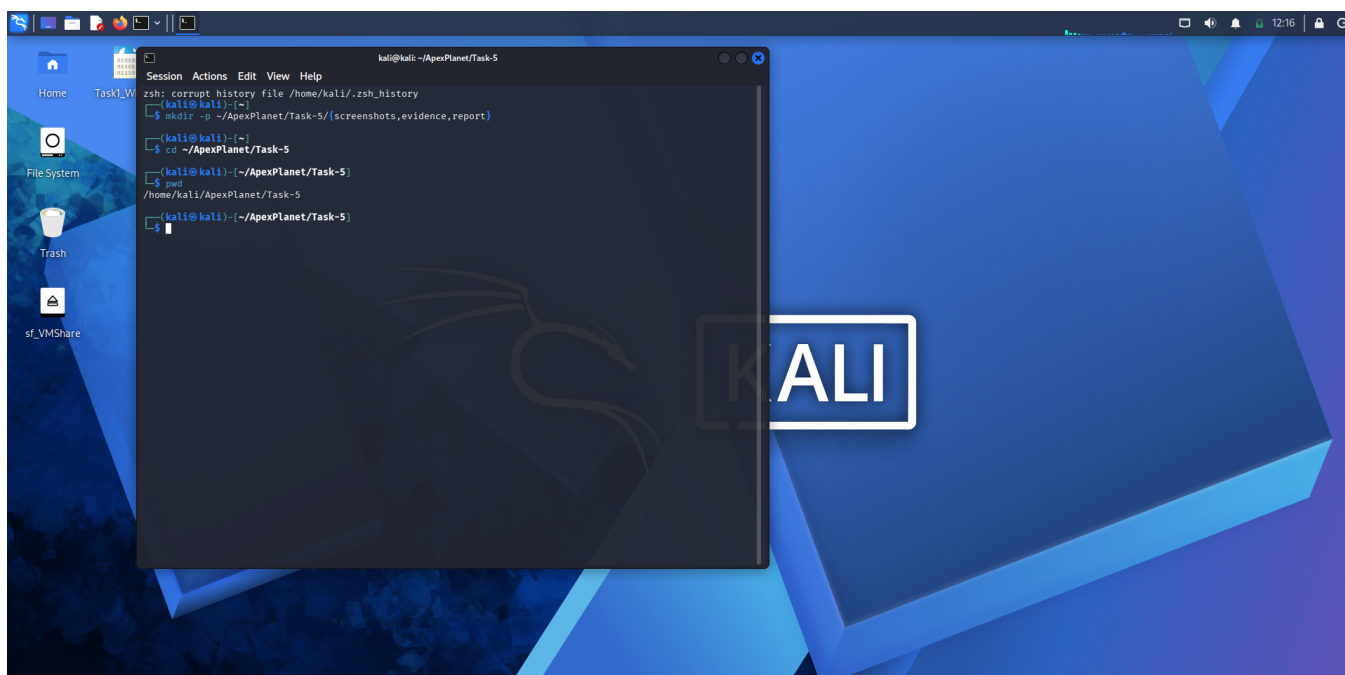
Disclaimer

This project was performed only in an authorized, isolated laboratory environment using DVWA for cybersecurity training and educational purposes. No unauthorized systems were targeted.

10. Evidence Appendix — Submitted Screenshots

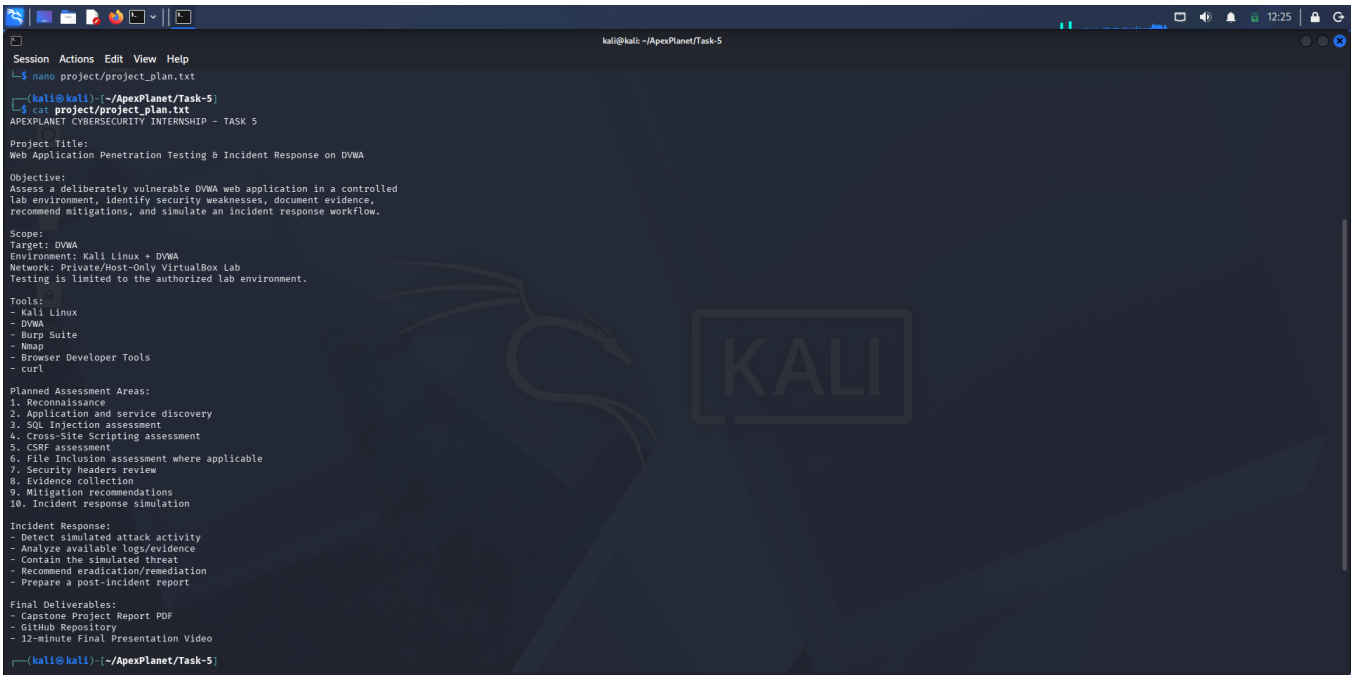
This appendix includes all 21 screenshots supplied in the Task-5 ZIP. Screenshots are retained as evidence and are captioned using their original filenames.

Evidence 1: 01-Task5-Project-Directory.png



01-Task5-Project-Directory.png

Evidence 2: 02-Task5-Project-Plan.png



```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help
└─$ nano project/project_plan.txt
(kali@kali) ~/ApexPlanet/Task-5
└─$ cat project/project_plan.txt
APEXPLANET CYBERSECURITY INTERNSHIP - TASK 5

Project Title:
Web Application Penetration Testing & Incident Response on DVWA

Objective:
Assess a deliberately vulnerable DVWA web application in a controlled
lab environment, identify security weaknesses, document evidence,
recommend mitigations, and simulate an incident response workflow.

Scope:
Target: DVWA
Environment: Kali Linux + DVWA
Network: Private/Host-Only VirtualBox Lab
Testing is limited to the authorized lab environment.

Tools:
- Kali Linux
- DVWA
- Burp Suite
- Nmap
- Browser Developer Tools
- curl

Planned Assessment Areas:
1. Reconnaissance
2. Application and service discovery
3. SQL Injection assessment
4. Cross-Site Scripting assessment
5. CSRF assessment
6. File Inclusion assessment where applicable
7. Security headers review
8. Evidence collection
9. Mitigation recommendations
10. Incident response simulation

Incident Response:
- Detect simulated attack activity
- Analyze available logs/evidence
- Contain the simulated threat
- Recommend eradication/remediation
- Prepare a post-incident report

Final Deliverables:
- Capstone Project Report PDF
- Github Repository
- 12-minute Final Presentation Video

(kali@kali) ~/ApexPlanet/Task-5
```

02-Task5-Project-Plan.png

Evidence 3: 03-DVWA-Apache-Running.png

```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help
- Contain the simulated threat
- Recommend eradication/remediation
- Prepare a post-incident report

Final Deliverables:
- Capstone Project Report PDF
- GitHub Repository
- 12-minute Final Presentation Video

(kali@kali)~(~/ApexPlanet/Task-5)
$ cd ~/ApexPlanet/Task-5
(kali@kali)~(~/ApexPlanet/Task-5)
$ ls -ld /var/www/html/DVWA
drwxr-xr-x 12 www-data www-data 4096 Aug 17 14:07 /var/www/html/DVWA

(kali@kali)~(~/ApexPlanet/Task-5)
$ sudo systemctl status apache2 --no-pager
[sudo] password for kali:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: https://httpd.apache.org/docs/2.4/

(kali@kali)~(~/ApexPlanet/Task-5)
$ sudo systemctl start apache2
(kali@kali)~(~/ApexPlanet/Task-5)
$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Tue 2020-09-15 12:28:39 EDT; 29s ago
     Invocation: d6849594a9ab0b9c4afe2aahf8a0
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 12547 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
   Tasks: 6 (limit: 4469)
   Memory: 23.3M (peak: 23.5M)
   CPU: 140ms
   CGroup: /system.slice/apache2.service
           └─12547 /usr/sbin/apache2 -k start -DFOREGROUND
           └─12564 /usr/sbin/apache2 -k start -DFOREGROUND
           └─12565 /usr/sbin/apache2 -k start -DFOREGROUND
           └─12566 /usr/sbin/apache2 -k start -DFOREGROUND
           └─12567 /usr/sbin/apache2 -k start -DFOREGROUND
           └─12569 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 15 12:28:39 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 15 12:28:39 kali apache2[12547]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:28:39 kali systemd[1]: Started apache2.service - The Apache HTTP Server.

(kali@kali)~(~/ApexPlanet/Task-5)
```

03-DVWA-Apache-Running.png

Evidence 4: 05-MariaDB-Running.png

```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help
[Tue Sep 15 12:28:39.419734 2026] [mpm_prefork:notice] [pid 12547:tid 12547] AH00163: Apache/2.4.68 (Debian) configured -- resuming normal operations
[Tue Sep 15 12:28:39.419771 2026] [core:notice] [pid 12547:tid 12547] AH00094: Command line: '/usr/sbin/apache2 -D FOREGROUND'
[Tue Sep 15 12:32:45.899297 2026] [php:error] [pid 12564:tid 12564] [client 127.0.0.1:53768] PHP Fatal error:  Uncaught mysqli_sql_exception: Connection refused in /var/www/html/DVWA/dwa/includes/dwaPage.inc.php:570\nStack trace:\n#0 /var/www/html/DVWA/dwa/includes/dwaPage.inc.php(570): mysqli_connect()\n#1 /var/www/html/DVWA/login.php(8): dwaDatabaseConnect()\n#2 {main}\n thrown in /var/www/html/DVWA/dwa/includes/dwaPage.inc.php on line 570
[Tue Sep 15 12:32:58.893245 2026] [php:error] [pid 12567:tid 12567] [client 127.0.0.1:58298] PHP Fatal error:  Uncaught mysqli_sql_exception: Connection refused in /var/www/html/DVWA/dwa/includes/dwaPage.inc.php:570\nStack trace:\n#0 /var/www/html/DVWA/dwa/includes/dwaPage.inc.php(570): mysqli_connect()\n#1 /var/www/html/DVWA/login.php(8): dwaDatabaseConnect()\n#2 {main}\n thrown in /var/www/html/DVWA/dwa/includes/dwaPage.inc.php on line 570
[Tue Sep 15 12:33:04.052166 2026] [php:error] [pid 12566:tid 12566] [client 127.0.0.1:58304] PHP Fatal error:  Uncaught mysqli_sql_exception: Connection refused in /var/www/html/DVWA/dwa/includes/dwaPage.inc.php:570\nStack trace:\n#0 /var/www/html/DVWA/dwa/includes/dwaPage.inc.php(570): mysqli_connect()\n#1 /var/www/html/DVWA/login.php(8): dwaDatabaseConnect()\n#2 {main}\n thrown in /var/www/html/DVWA/dwa/includes/dwaPage.inc.php on line 570

kali@kali:~/ApexPlanet/Task-5
$ sudo systemctl status mariadb --no-pager
o mariadb.service - MariaDB 11.8.8 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/

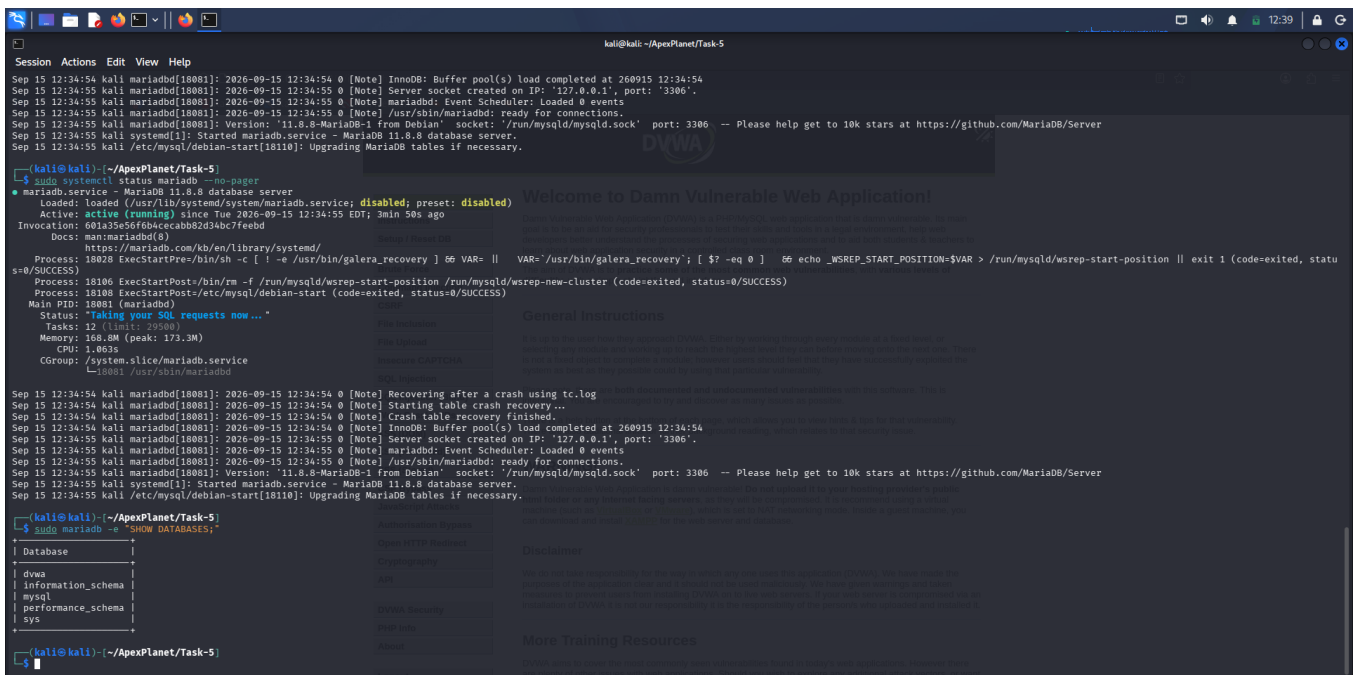
kali@kali:~/ApexPlanet/Task-5
$ sudo systemctl start mariadb

kali@kali:~/ApexPlanet/Task-5
$ sudo systemctl status mariadb --no-pager
o mariadb.service - MariaDB 11.8.8 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 12:34:55 EDT; 9s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
  Process: 18028 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= /usr/bin/galera_recovery'; [ $? -eq 0 ] && echo _WSREP_START_POSITION=$VAR > /run/mysqld/wsrep-start-position || exit 1 (code=exited, status=0/SUCCESS)
  Process: 18106 ExecStartPost=/bin/rm -f /run/mysqld/wsrep-start-position /run/mysqld/wsrep-new-cluster (code=exited, status=0/SUCCESS)
  Process: 18188 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
 Main PID: 18081 (mariadb)
   Tasks: 16 (limit: 29580)
  Memory: 168.7M (peak: 173.3M)
    CPU: 947ms
   CGroup: /system.slice/mariadb.service
           └─18081 /usr/sbin/mariadb

Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Recovering after a crash using tc.log
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Starting table crash recovery...
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Crash table recovery finished.
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] InnoDB: Buffer pool(s) load completed at 260915 12:34:54
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] mariadb: Event Scheduler: Loaded 0 events
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] /usr/sbin/mariadb: ready for connections.
Sep 15 12:34:55 kali mariadb[18081]: Version: '11.8.8-MariaDB-1 from Debian' socket: '/run/mysqld/mysqld.sock' port: 3306 -- Please help get to 10k stars at https://github.com/MariaDB/Server
Sep 15 12:34:55 kali systemd[1]: Started mariadb.service - MariaDB 11.8.8 database server.
Sep 15 12:34:55 kali /etc/mysql/debian-start[18110]: Upgrading MariaDB tables if necessary.
```

05-MariaDB-Running.png

Evidence 5: 06-DVWA-Database-Verified.png



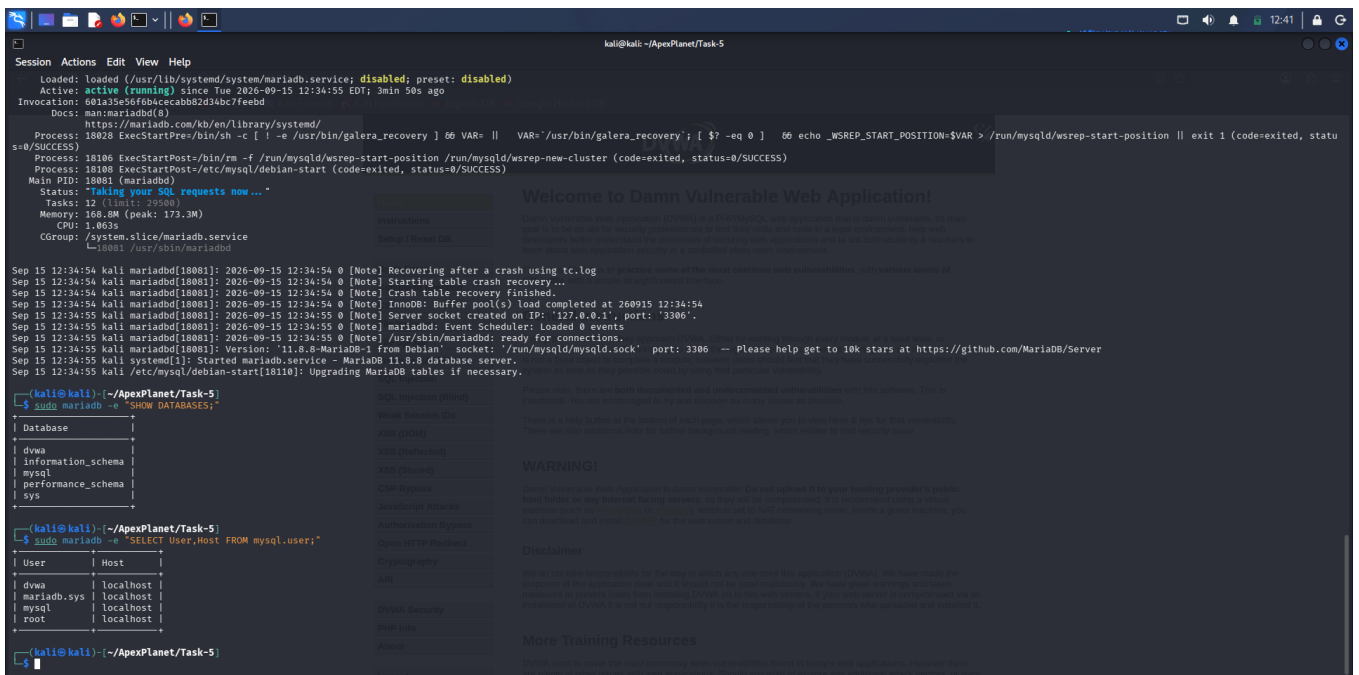
```
kali@kali: ~/ApexPlanet/Task-5
$ sudo systemctl status mariadb --no-pager
● mariadb.service - MariaDB 11.8.8 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 12:34:55 EDT; 3min 58s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 18028 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= || VAR=/usr/bin/galera_recovery; [ $? -eq 0 ] && echo _WSREP_START_POSITION=$VAR > /run/mysqld/wsrep-start-position || exit 1 (code=exited, status=0/SUCCESS)
   Process: 18106 ExecStartPost=/bin/rm -f /run/mysqld/wsrep-start-position /run/mysqld/wsrep-new-cluster (code=exited, status=0/SUCCESS)
   Process: 18108 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
   Main PID: 18081 (mariadb)
     Status: "taking your SQL requests now..."
       Tasks: 12 (limit: 4915)
      Memory: 168.8M (peak: 173.3M)
         CPU: 1.063s
        CGroup: /system.slice/mariadb.service
                └─18081 /usr/sbin/mariadb

Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] InnoDB: Buffer pool(s) load completed at 260915 12:34:54
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] mariadb: Event Scheduler: Loaded 0 events
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] /usr/sbin/mariadb: ready for connections.
Sep 15 12:34:55 kali mariadb[18081]: Version: '11.8.8-MariaDB-1 from Debian' socket: '/run/mysqld/mysqld.sock' port: 3306 -- Please help get to 10k stars at https://github.com/MariaDB/Server
Sep 15 12:34:55 kali system[1]: Started mariadb.service - MariaDB 11.8.8 database server.
Sep 15 12:34:55 kali /etc/mysql/debian-start[18110]: Upgrading MariaDB tables if necessary.

kali@kali:~/ApexPlanet/Task-5
$ sudo mariadb -e "SHOW DATABASES;"
+-----+
| Database |
+-----+
| dwva      |
| information_schema |
| mysql     |
| performance_schema |
| sys       |
+-----+
```

06-DVWA-Database-Verified.png

Evidence 6: 07-DVWA-Database-User.png



07-DVWA-Database-User.png

Evidence 7: 08-DVWA-Database-Grants.png

```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help
Main PID: 18081 (mariadb)
Status: "Taking your SQL requests now..."
Tasks: 12 (limit: 9999)
Memory: 168.8M (peak: 173.3M)
CPU: 1.063s
CGroup: /system.slice/mariadb.service
        └─18081 /usr/sbin/mariadb

Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Recovering after a crash using tc.log
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Starting table crash recovery...
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Crash table recovery finished.
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] InnoDB: Buffer pool(s) load completed at 260915 12:34:54
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] Server socket created on IP: '127.0.0.1' port: '3306'
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] mariadb: Event Scheduler: Loaded 0 events
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] /usr/sbin/mariadb: ready for connections.
Sep 15 12:34:55 kali mariadb[18081]: Version: '11.8.8-MariaDB-1 from Debian' sockets: /run/mysqld/mysqld.sock port: 3306 -- Please help get to 10k stars at https://github.com/MariaDB/Server
Sep 15 12:34:55 kali systemd[1]: Started mariadb.service - MariaDB 11.8.8 database server.
Sep 15 12:34:55 kali /etc/mysql/debian-start[18110]: Upgrading MariaDB tables if necessary. A lot of things in its practice some of the most common web vulnerabilities, with various levels of
complexity, with a simple management interface.

(kali@kali): ~/ApexPlanet/Task-5
$ sudo mariadb -e "SHOW DATABASES;"
+-----+
| Database |
+-----+
| dwwa     |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+

(kali@kali): ~/ApexPlanet/Task-5
$ sudo mariadb -e "SELECT User,Host FROM mysql.user;"
+-----+-----+
| User | Host |
+-----+-----+
| dwwa  | localhost |
| mariadb.sys | localhost |
| mysql | localhost |
| root  | localhost |
+-----+-----+

(kali@kali): ~/ApexPlanet/Task-5
$ sudo mariadb -e "SHOW GRANTS FOR 'dwwa'@'localhost';"
GRANTS for dwwa@localhost
GRANT USAGE ON *.* TO 'dwwa'@'localhost' IDENTIFIED BY PASSWORD '*d67b38cdd1a55623e05f55856a29b9654ff823d'
GRANT ALL PRIVILEGES ON 'dwwa'.* TO 'dwwa'@'localhost'

(kali@kali): ~/ApexPlanet/Task-5
```

08-DVWA-Database-Grants.png

Evidence 8: 09-DVWA-Database-Config.png

```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Starting table crash recovery ...
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] Crash table recovery finished.
Sep 15 12:34:54 kali mariadb[18081]: 2026-09-15 12:34:54 0 [Note] InnoDB: Buffer pool(s) load completed at 260915 12:34:54
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] mariadb: Event Scheduler: Loaded 0 events
Sep 15 12:34:55 kali mariadb[18081]: 2026-09-15 12:34:55 0 [Note] /usr/sbin/mariadbd: ready for connections.
Sep 15 12:34:55 kali mariadb[18081]: Version: '11.8.8-MariaDB-1 from Debian' socket: '/run/mysqld/mysqld.sock' port: 3306 -- Please help get to 10k stars at https://github.com/MariaDB/Server
Sep 15 12:34:55 kali systemd[1]: Started mariadb.service - MariaDB 11.8.8 database server.
Sep 15 12:34:55 kali /etc/mysql/debian-start[18110]: Upgrading MariaDB tables if necessary.

(kali@kali)~(~/ApexPlanet/Task-5)
$ sudo mariadb -e "SHOW DATABASES;"
+-----+
| Database |
+-----+
| dwva     |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+

(kali@kali)~(~/ApexPlanet/Task-5)
$ sudo mariadb -e "SELECT User,Host FROM mysql.user;"
+-----+-----+
| User | Host |
+-----+-----+
| dwva | localhost |
| mariadb.sys | localhost |
| mysql | localhost |
| root | localhost |
+-----+-----+

(kali@kali)~(~/ApexPlanet/Task-5)
$ sudo mariadb -e "SHOW GRANTS FOR 'dwva'@'localhost';"
+-----+
| Grants for dwva@localhost |
+-----+
| GRANT USAGE ON *.* TO 'dwva'@'localhost' IDENTIFIED BY PASSWORD '*D67B38C0CD1A55623ED5F5856A29B9654FF823D' |
| GRANT ALL PRIVILEGES ON 'dwva'.* TO 'dwva'@'localhost' |
+-----+

(kali@kali)~(~/ApexPlanet/Task-5)
$ grep -nE "db_server|db_database|db_user|db_password" /var/www/html/DVWA/config/config.inc.php
12: try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
18: $DWVA[ 'db_server' ] = getenv( 'DB_SERVER' ) ? : '127.0.0.1';
19: $DWVA[ 'db_database' ] = getenv( 'DB_DATABASE' ) ? : 'dwva';
20: $DWVA[ 'db_user' ] = getenv( 'DB_USER' ) ? : 'dwva';
21: $DWVA[ 'db_password' ] = getenv( 'DB_PASSWORD' ) ? : 'dwva';

Welcome to Damn Vulnerable Web Application!

Damn Vulnerable Web Application (DVWA) is a PHP/MySQL web application that is secure (when installed & configured correctly), but is also intentionally insecure. The aim of DVWA is to provide some of the most common web vulnerabilities, with various levels of difficulty, with a simple user-friendly interface.

General Instructions

It is up to the user how they approach DVWA. Either by working through every module as a final goal, or selecting any module and working up to reach the highest level they can before moving onto the next one. There is no set order to complete a module, however users should bear that they have successfully completed the system as long as they provide valid by using first particular instructions.

Please note, there are both documented and undocumented vulnerabilities with this software. This is intentional. You are encouraged to try and discover as many vulnerabilities as possible.

There is a hint button at the bottom of each page, which allows you to view hints & tips for that vulnerability. There are also references from the further reading/reading, which relates to that security module.

More Training Resources
```

09-DVWA-Database-Config.png

Evidence 9: 10-DVWA-Database-Connection.png

```
kali@kali: ~/ApexPlanet/Task-5
mysql
performance_schema
sys

(kali@kali)~$ sudo mariadb -e "SELECT User,Host FROM mysql.user;"
+-----+-----+
| User | Host |
+-----+-----+
| dwva | localhost |
| mariadb.sys | localhost |
| mysql | localhost |
| root | localhost |
+-----+-----+

(kali@kali)~$ sudo mariadb -e "SHOW GRANTS FOR 'dwva'@'localhost';"
+-----+-----+
| Grants for dwva@localhost |
+-----+-----+
| GRANT USAGE ON *.* TO 'dwva'@'localhost' IDENTIFIED BY PASSWORD '*D67B38CDD1A55623ED9F55856A29B9654FF823D' |
| GRANT ALL PRIVILEGES ON 'dwva'.* TO 'dwva'@'localhost' |
+-----+-----+

(kali@kali)~$ grep -nE "db_server|db_database|db_user|db_password" /var/www/html/DVWA/config/config.inc.php
4: try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
322 WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
18:$ DWVA['db_server'] = getenv('DB_SERVER') ? '127.0.0.1';
19:$ DWVA['db_database'] = getenv('DB_DATABASE') ? 'dwva';
20:$ DWVA['db_user'] = getenv('DB_USER') ? 'dwva';
21:$ DWVA['db_password'] = getenv('DB_PASSWORD') ? 'dwva';

(kali@kali)~$ mysql -h 127.0.0.1 -u dwva -p dwva
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 20
Server version: 11.6.3-MariaDB-1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

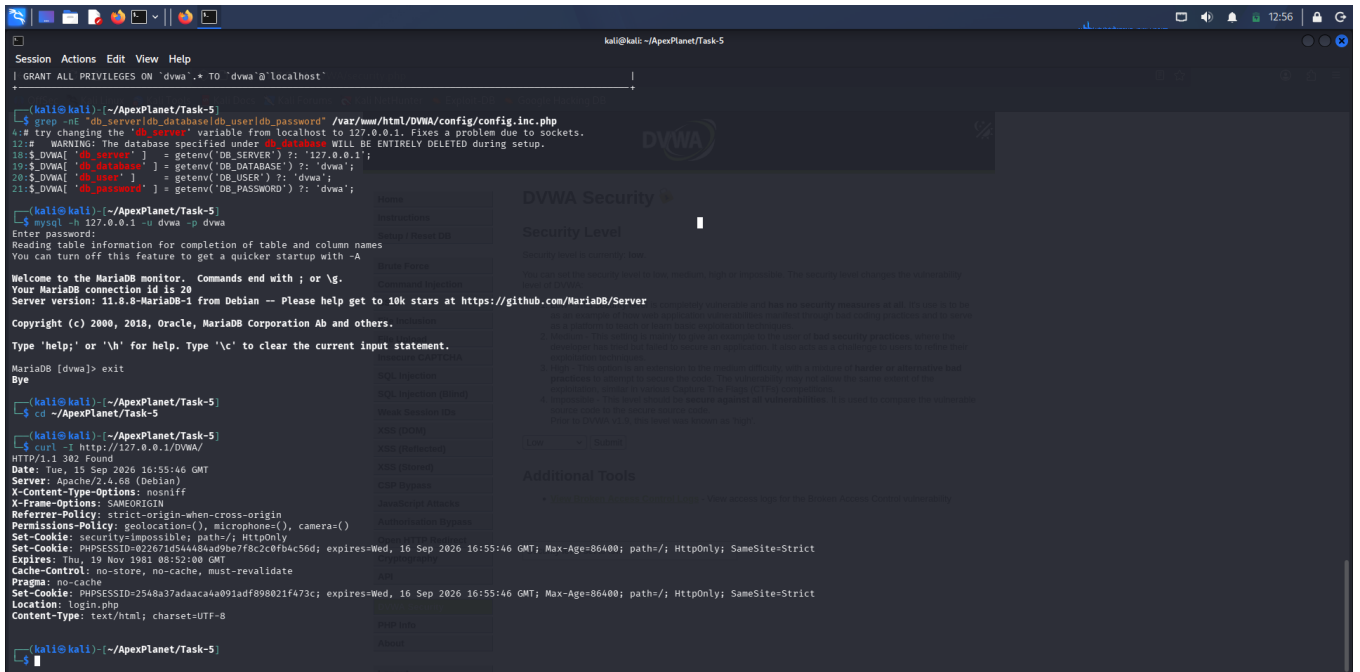
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [dwva]> exit
Bye

(kali@kali)~$
```

10-DVWA-Database-Connection.png

Evidence 10: 11-DVWA-HTTP-Headers.png



11-DVWA-HTTP-Headers.png

Evidence 11: 12-DVWA-Recon-Web-Database.png

The screenshot displays a Kali Linux terminal window and the DVWA (Damn Vulnerable Web Application) web interface. The terminal shows the execution of an Nmap scan on the local host (127.0.0.1), identifying the Apache Web Server and the DVWA application. The DVWA interface shows the 'Vulnerability: SQL Injection' page, which displays the results of the scan, including the discovered service (Apache/2.4.68) and the detected vulnerability (SQL Injection).

Terminal Output:

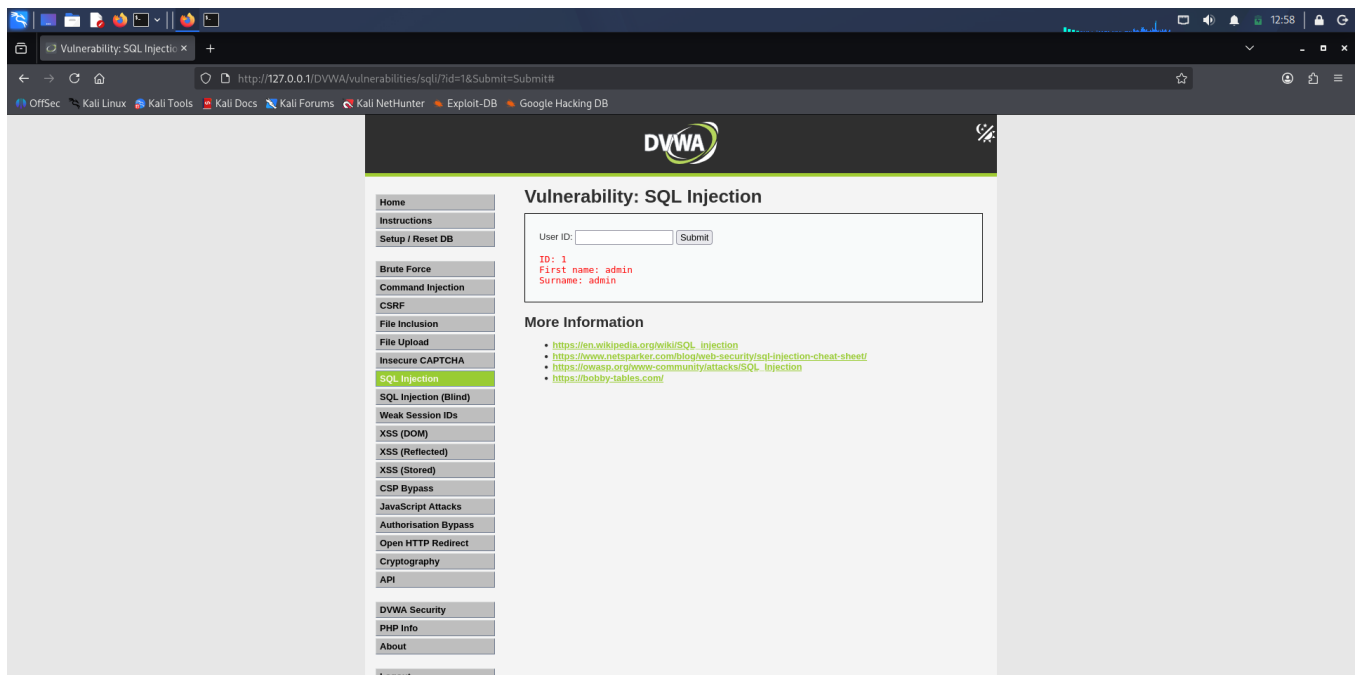
```
kali@kali: ~/ApexPlanet/Task-5
$ cd ~/ApexPlanet/Task-5
$ nmap -sV -p 80,3306 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-15 13:08 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000063s latency).
PORT      STATE SERVICE
80/tcp    open  http
3306/tcp  open  mysql
Service detected on 3306/tcp: MySQL 5.7.44-log
Service detected on 80/tcp: Apache/2.4.68 ((Debian))
Nmap done: 1 IP address (1 host up) scanned in 16.66 seconds
```

DVWA Interface:

The DVWA interface shows the 'Vulnerability: SQL Injection' page. The 'Vulnerability' section displays the results of the scan, including the discovered service (Apache/2.4.68) and the detected vulnerability (SQL Injection). The 'More Information' section provides additional details about the vulnerability, including a link to the Nmap service fingerprint and a link to the CVE database.

12-DVWA-Recon-Web-Database.png

Evidence 12: 12-SQLi-Baseline.png



12-SQLi-Baseline.png

Evidence 13: 13-DVWA-Web-Headers.png

The screenshot shows a Kali Linux terminal window with the following commands and output:

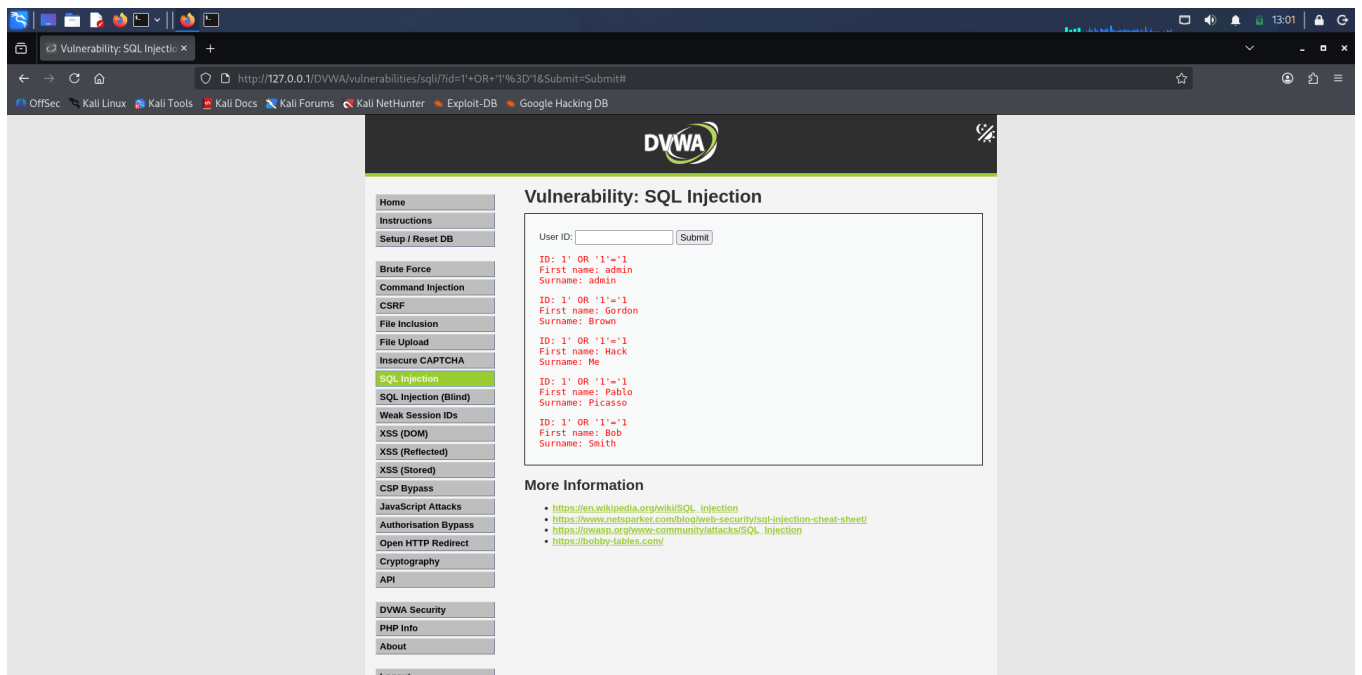
```
kali@kali: ~/ApexPlanet/Task-5
$ cd ~/ApexPlanet/Task-5
$ curl -I http://127.0.0.1/DVWA/ | tee web_headers.txt
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 17:10:30 GMT
Server: Apache/2.4.68 (Debian)
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referrer-Policy: strict-origin-when-cross-origin
Permissions-Policy: geolocation(), microphone(), camera=()
Set-Cookie: security=impossible; path=/; HttpOnly
Set-Cookie: PHPSESSID=ee28e81981a8d0267a414823b8c79f40; expires=Wed, 16 Sep 2026 17:10:30 GMT; Max-Age=86400; path=/; HttpOnly; SameSite=Strict
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Set-Cookie: PHPSESSID=77c2c82df986d65cb9523f175b34b0c; expires=Wed, 16 Sep 2026 17:10:30 GMT; Max-Age=86400; path=/; HttpOnly; SameSite=Strict
Location: login.php
Content-Type: text/html; charset=UTF-8

$ cat web_headers.txt
HTTP/1.1 302 Found
Date: Tue, 15 Sep 2026 17:10:30 GMT
Server: Apache/2.4.68 (Debian)
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Referrer-Policy: strict-origin-when-cross-origin
Permissions-Policy: geolocation(), microphone(), camera=()
Set-Cookie: security=impossible; path=/; HttpOnly
Set-Cookie: PHPSESSID=ee28e81981a8d0267a414823b8c79f40; expires=Wed, 16 Sep 2026 17:10:30 GMT; Max-Age=86400; path=/; HttpOnly; SameSite=Strict
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Set-Cookie: PHPSESSID=77c2c82df986d65cb9523f175b34b0c; expires=Wed, 16 Sep 2026 17:10:30 GMT; Max-Age=86400; path=/; HttpOnly; SameSite=Strict
Location: login.php
Content-Type: text/html; charset=UTF-8
```

The DVWA web application interface is visible in the background, showing the "Vulnerability: SQL Injection" section. The "More Information" section lists several links for further reading on SQL injection attacks.

13-DVWA-Web-Headers.png

Evidence 14: 13-SQLi-Vulnerability-Confirmed.png



13-SQLi-Vulnerability-Confirmed.png

Evidence 15: 14-Apache-Access-Log-Baseline.png

```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help

127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /evox/about HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.1" 200 11132 "-"
127.0.0.1 - - [15/Sep/2026:13:10:30 -0400] "HEAD /DVWA/ HTTP/1.1" 302 777 "-" curl/8.20.0"

kali@kali:~/ApexPlanet/Task-5
$ sudo tail -n 20 /var/log/apache2/access.log | tee -a ApexPlanet/Task-5/apache_access_baseline.txt
127.0.0.1 - - [15/Sep/2026:12:50:03 -0400] "GET /DVWA/index.php HTTP/1.1" 200 2974 "http://127.0.0.1/DVWA/login.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:50:10 -0400] "GET /DVWA/ HTTP/1.1" 200 2974 "-" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:03 -0400] "GET /DVWA/security.php HTTP/1.1" 200 2431 "http://127.0.0.1/DVWA/index.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:04 -0400] "GET /DVWA/dvwa/images/lock.png HTTP/1.1" 304 422 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:18 -0400] "POST /DVWA/security.php HTTP/1.1" 302 673 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:18 -0400] "GET /DVWA/security.php HTTP/1.1" 200 2443 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:55:46 -0400] "HEAD /DVWA/ HTTP/1.1" 302 777 "-" curl/8.20.0"
127.0.0.1 - - [15/Sep/2026:12:57:52 -0400] "GET /DVWA/vulnerabilities/sql/ HTTP/1.1" 200 1963 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:58:44 -0400] "GET /DVWA/vulnerabilities/sql/1?id=16Submit-Submit HTTP/1.1" 200 1986 "http://127.0.0.1/DVWA/vulnerabilities/sql/" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:13:01:04 -0400] "GET /DVWA/vulnerabilities/sql/1?id=1527-OR+92715273302715Submit-Submit HTTP/1.1" 200 2044 "http://127.0.0.1/DVWA/vulnerabilities/sql/1?id=16Submit-Submit" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:13:08:10 -0400] "GET / HTTP/1.0" 200 11151 "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "POST /sdx HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /mmaplowercheck1789492101 HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /HMAP1 HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /evox/about HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.1" 200 11132 "-"
127.0.0.1 - - [15/Sep/2026:13:10:30 -0400] "HEAD /DVWA/ HTTP/1.1" 302 777 "-" curl/8.20.0"

kali@kali:~/ApexPlanet/Task-5
$ cat -a ApexPlanet/Task-5/apache_access_baseline.txt
127.0.0.1 - - [15/Sep/2026:12:50:03 -0400] "GET /DVWA/index.php HTTP/1.1" 200 2974 "http://127.0.0.1/DVWA/login.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:50:10 -0400] "GET /DVWA/ HTTP/1.1" 200 2974 "-" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:03 -0400] "GET /DVWA/security.php HTTP/1.1" 200 2431 "http://127.0.0.1/DVWA/index.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:04 -0400] "GET /DVWA/dvwa/images/lock.png HTTP/1.1" 304 422 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:18 -0400] "POST /DVWA/security.php HTTP/1.1" 302 673 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:52:18 -0400] "GET /DVWA/security.php HTTP/1.1" 200 2443 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:55:46 -0400] "HEAD /DVWA/ HTTP/1.1" 302 777 "-" curl/8.20.0"
127.0.0.1 - - [15/Sep/2026:12:57:52 -0400] "GET /DVWA/vulnerabilities/sql/ HTTP/1.1" 200 1963 "http://127.0.0.1/DVWA/security.php" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:12:58:44 -0400] "GET /DVWA/vulnerabilities/sql/1?id=16Submit-Submit HTTP/1.1" 200 1986 "http://127.0.0.1/DVWA/vulnerabilities/sql/" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:13:01:04 -0400] "GET /DVWA/vulnerabilities/sql/1?id=1527-OR+92715273302715Submit-Submit HTTP/1.1" 200 2044 "http://127.0.0.1/DVWA/vulnerabilities/sql/1?id=16Submit-Submit" Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - - [15/Sep/2026:13:08:10 -0400] "GET / HTTP/1.0" 200 11151 "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "POST /sdx HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /mmaplowercheck1789492101 HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /HMAP1 HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET /evox/about HTTP/1.1" 404 665 "-" Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-"
127.0.0.1 - - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.1" 200 11132 "-"
127.0.0.1 - - [15/Sep/2026:13:10:30 -0400] "HEAD /DVWA/ HTTP/1.1" 302 777 "-" curl/8.20.0"
```

14-Apache-Access-Log-Baseline.png

Evidence 16: 15-SQLi-Log-Indicator.png

```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help

127.0.0.1 - [15/Sep/2026:12:57:52 -0400] "GET /DWA/vulnerabilities/sqli/ HTTP/1.1" 200 1963 "http://127.0.0.1/DWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:58:44 -0400] "GET /DWA/vulnerabilities/sqli/?id=16Submit-Submit HTTP/1.1" 200 1986 "http://127.0.0.1/DWA/vulnerabilities/sqli/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:13:01:04 -0400] "GET /DWA/vulnerabilities/sqli/?id=1&2+OR+&271&273&2716Submit-Submit HTTP/1.1" 200 2044 "http://127.0.0.1/DWA/vulnerabilities/sqli/?id=16Submit-Submit" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:13:08:10 -0400] "GET / HTTP/1.0" 200 11151 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "POST /sdk HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /mmaplowercheck1789492101 HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-" "-"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /HMAPI HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /vbox/about HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-" "-"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.1" 200 11132 "-" "-"
127.0.0.1 - [15/Sep/2026:13:10:30 -0400] "HEAD /DWA/ HTTP/1.1" 302 777 "-" "curl/8.20.0"

kali@kali:~/ApexPlanet/Task-5
$ cat ~/ApexPlanet/Task-5/apache_access_baseline.txt
127.0.0.1 - [15/Sep/2026:12:50:03 -0400] "GET /DWA/index.php HTTP/1.1" 200 2974 "http://127.0.0.1/DWA/login.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:50:10 -0400] "GET /DWA/ HTTP/1.1" 200 2974 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:52:03 -0400] "GET /DWA/security.php HTTP/1.1" 200 2431 "http://127.0.0.1/DWA/index.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:52:04 -0400] "GET /DWA/dwa/images/lock.png HTTP/1.1" 304 622 "http://127.0.0.1/DWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:52:18 -0400] "POST /DWA/security.php HTTP/1.1" 302 673 "http://127.0.0.1/DWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:52:18 -0400] "GET /DWA/security.php HTTP/1.1" 200 2443 "http://127.0.0.1/DWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:55:16 -0400] "HEAD /DWA/ HTTP/1.1" 302 777 "-" "curl/8.20.0"
127.0.0.1 - [15/Sep/2026:12:57:52 -0400] "GET /DWA/vulnerabilities/sqli/ HTTP/1.1" 200 1963 "http://127.0.0.1/DWA/security.php" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:12:58:44 -0400] "GET /DWA/vulnerabilities/sqli/?id=16Submit-Submit HTTP/1.1" 200 1986 "http://127.0.0.1/DWA/vulnerabilities/sqli/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:13:01:04 -0400] "GET /DWA/vulnerabilities/sqli/?id=1&2+OR+&271&273&2716Submit-Submit HTTP/1.1" 200 2044 "http://127.0.0.1/DWA/vulnerabilities/sqli/?id=16Submit-Submit" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:13:08:10 -0400] "GET / HTTP/1.0" 200 11151 "-" "-"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "POST /sdk HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /mmaplowercheck1789492101 HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11151 "-" "-"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /HMAPI HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /vbox/about HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET / HTTP/1.0" 200 11132 "-" "-"
127.0.0.1 - [15/Sep/2026:13:10:30 -0400] "HEAD /DWA/ HTTP/1.1" 302 777 "-" "curl/8.20.0"

kali@kali:~/ApexPlanet/Task-5
$ cd ~/ApexPlanet/Task-5
kali@kali:~/ApexPlanet/Task-5
$ grep -Ei "sqli&271&273&2716Submit-Submit" apache_access_baseline.txt
127.0.0.1 - [15/Sep/2026:12:58:44 -0400] "GET /DWA/vulnerabilities/sqli/?id=16Submit-Submit HTTP/1.1" 200 1986 "http://127.0.0.1/DWA/vulnerabilities/sqli/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:13:01:04 -0400] "GET /DWA/vulnerabilities/sqli/?id=1&2+OR+&271&273&2716Submit-Submit HTTP/1.1" 200 2044 "http://127.0.0.1/DWA/vulnerabilities/sqli/?id=16Submit-Submit" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "POST /sdk HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /mmaplowercheck1789492101 HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /HMAPI HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
127.0.0.1 - [15/Sep/2026:13:08:21 -0400] "GET /vbox/about HTTP/1.1" 404 665 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
```

15-SQLi-Log-Indicator.png

Evidence 17: 16-Incident-Containment-Apache-Stopped.png

```
kali@kali: ~/ApexPlanet/Task-5
$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 12:28:39 EDT; 49min ago
  Invocation: d6884959494b02b9c8a4fe2aabf9a6
    Docs: https://httpd.apache.org/docs/2.4/
  Process: 28464 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
 Main PID: 12547 (apache2)
   Status: "Total requests: 40; Idle/Busy workers 100/0; Requests/sec: 0.0136; Bytes served/sec: 35 B/sec"
    Tasks: 7 (limit: 4469)
  Memory: 28.8M (peak: 40.1M)
    CPU: 834ms
   CGroup: /system.slice/apache2.service
           └─12547 /usr/sbin/apache2 -k start -DFOREGROUND
             └─28471 /usr/sbin/apache2 -k start -DFOREGROUND
               └─28473 /usr/sbin/apache2 -k start -DFOREGROUND
                 └─28474 /usr/sbin/apache2 -k start -DFOREGROUND
                   └─28475 /usr/sbin/apache2 -k start -DFOREGROUND
                     └─28476 /usr/sbin/apache2 -k start -DFOREGROUND
                       └─30570 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 15 12:28:39 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 15 12:28:39 kali apachectl[12547]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:28:39 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 15 12:47:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 12:47:27 kali apachectl[28464]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:47:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.

kali@kali:~/ApexPlanet/Task-5
$ sudo systemctl stop apache2

kali@kali:~/ApexPlanet/Task-5
$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
    Docs: https://httpd.apache.org/docs/2.4/

Sep 15 12:28:39 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 15 12:28:39 kali apachectl[12547]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:28:39 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 15 12:47:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 12:47:27 kali apachectl[28464]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:47:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.
Sep 15 13:18:04 kali systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Sep 15 13:18:04 kali apachectl[50186]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 13:18:04 kali systemd[1]: apache2.service: Deactivated successfully.
Sep 15 13:18:04 kali systemd[1]: Stopped apache2.service - The Apache HTTP Server.

kali@kali:~/ApexPlanet/Task-5
```

16-Incident-Containment-Apache-Stopped.png

Evidence 18: 17-Incident-Recovery-Apache-Restored.png

```
kali@kali: ~/ApexPlanet/Task-5
Session Actions Edit View Help
Sep 15 12:47:27 kali apache2[128464]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:47:27 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.

(kali@kali)~$ sudo systemctl stop apache2

(kali@kali)~$ sudo systemctl status apache2 --no-pager
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: https://httpd.apache.org/docs/2.4/

Sep 15 12:28:39 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 15 12:28:39 kali apache2[125471]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:28:39 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 15 12:47:27 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Sep 15 12:47:27 kali apache2[128464]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 12:47:27 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 15 13:18:04 kali systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Sep 15 13:18:04 kali apache2[50186]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 13:18:04 kali systemd[1]: apache2.service: Deactivated successfully.
Sep 15 13:18:04 kali systemd[1]: Stopped apache2.service - The Apache HTTP Server.

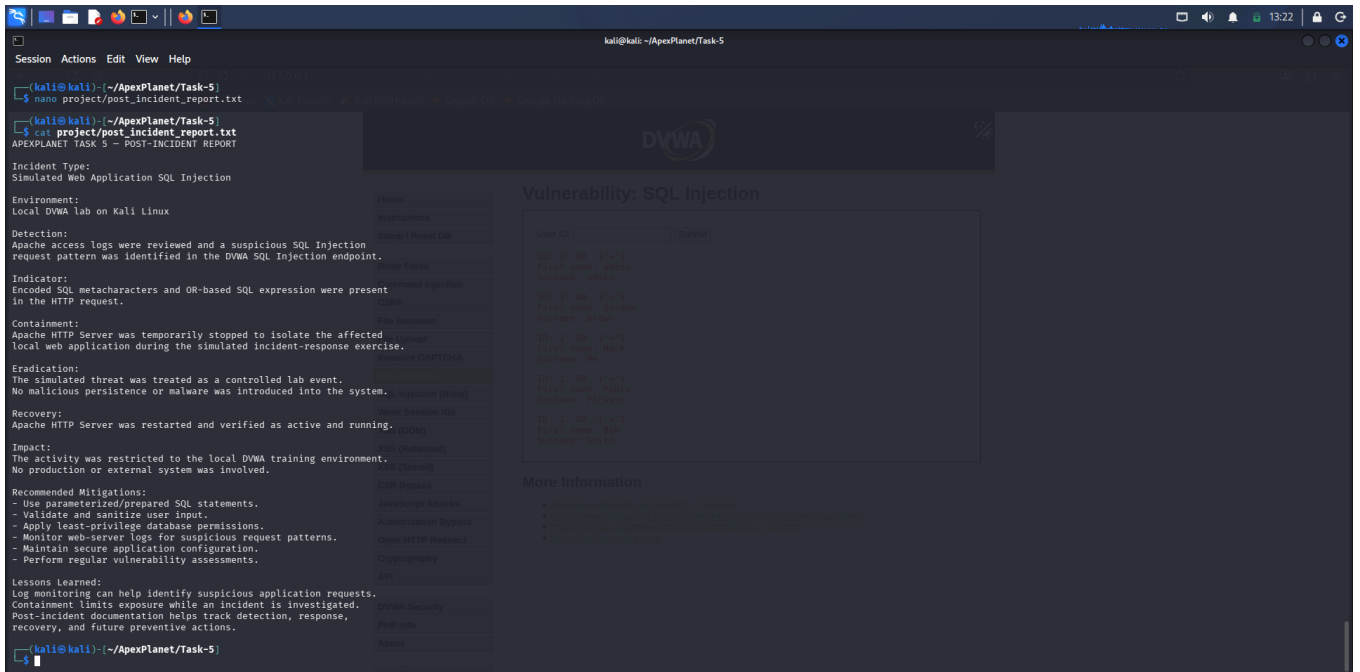
(kali@kali)~$ sudo systemctl start apache2
* apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 13:19:41 EDT; 3s ago
   Invocation: 6ba6c743c6455a0ab8ef6cfd1409
   Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 51732 (apache2)
   Status: "Processing requests ..."
   Tasks: 6 (limit: 4469)
   Memory: 15.4M (peak: 15.7M)
   CPU: 90ms
   CGroup: /system.slice/apache2.service
           └─51732 /usr/sbin/apache2 -k start -DFOREGROUND
           └─51735 /usr/sbin/apache2 -k start -DFOREGROUND
           └─51736 /usr/sbin/apache2 -k start -DFOREGROUND
           └─51737 /usr/sbin/apache2 -k start -DFOREGROUND
           └─51738 /usr/sbin/apache2 -k start -DFOREGROUND
           └─51748 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 15 13:19:41 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 15 13:19:41 kali apache2[51722]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Sep 15 13:19:41 kali systemd[1]: Started apache2.service - The Apache HTTP Server.

(kali@kali)~$
```

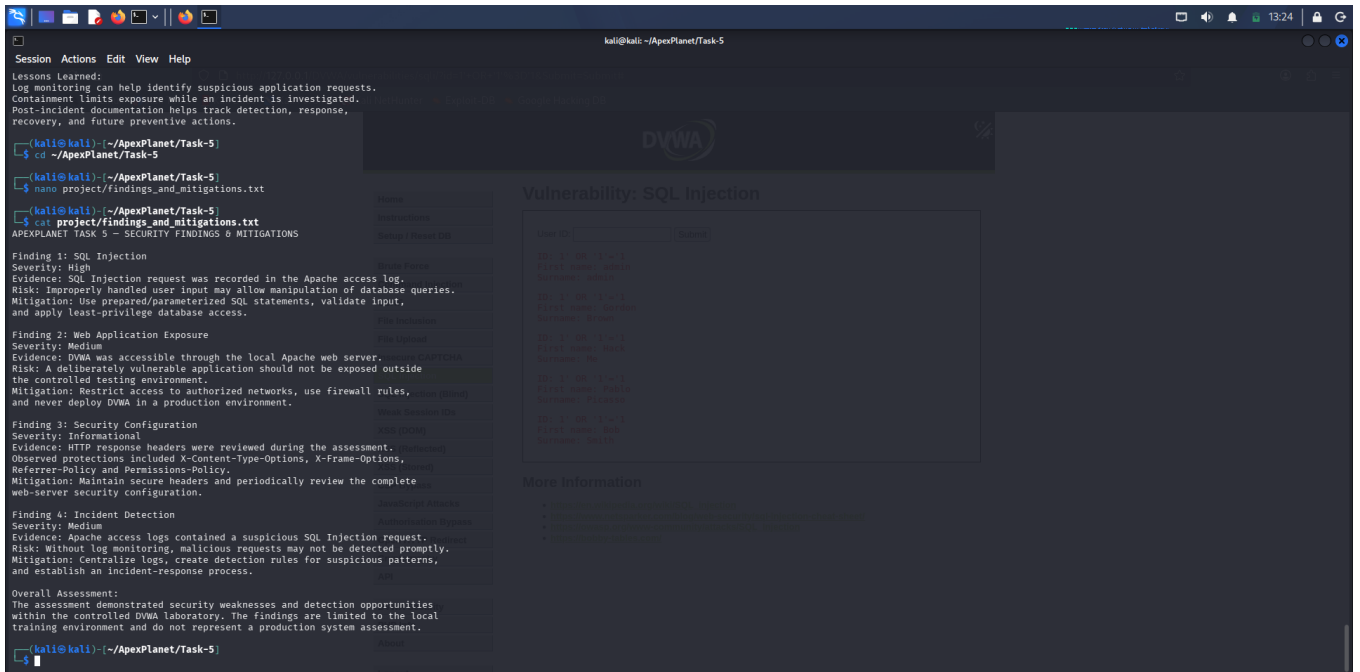
17-Incident-Recovery-Apache-Restored.png

Evidence 19: 18-Post-Incident-Report.png



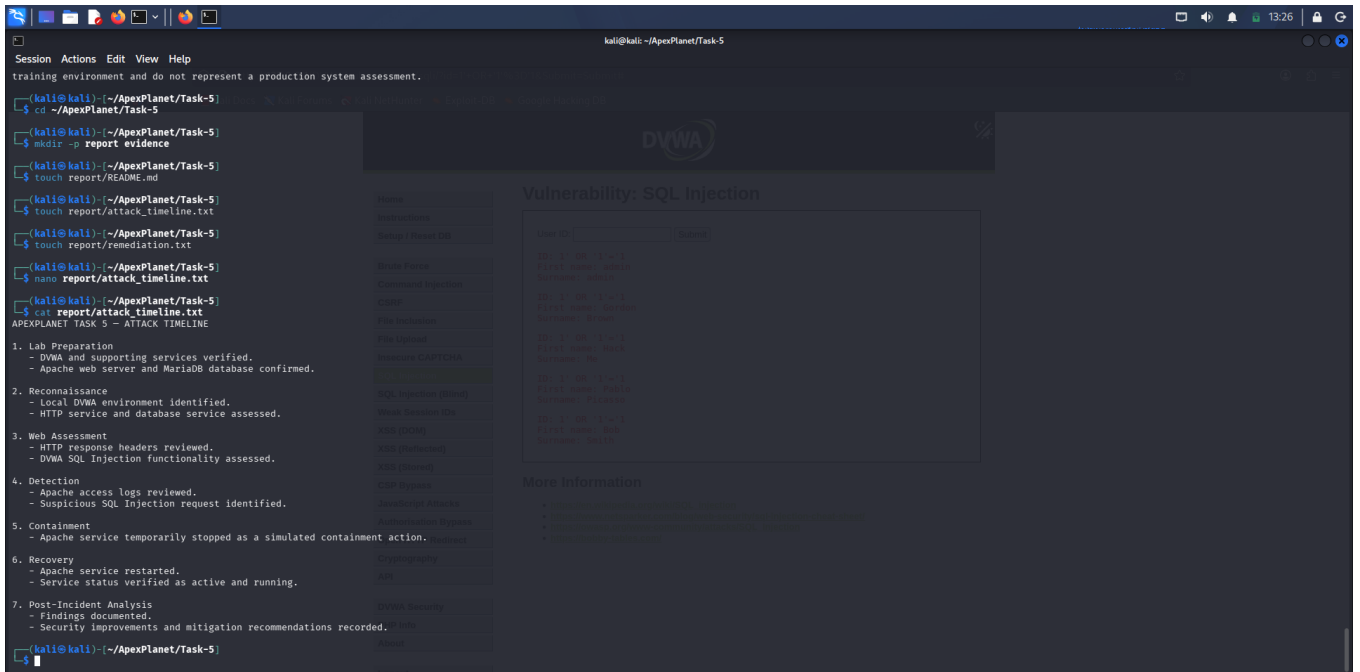
18-Post-Incident-Report.png

Evidence 20: 19-Findings-and-Mitigation.png



19-Findings-and-Mitigation.png

Evidence 21: 20-Attack-Timeline.png



20-Attack-Timeline.png

11. Conclusion

The Task-5 project evidence demonstrates a controlled DVWA security assessment and an associated incident-response simulation. The submitted artifacts document the lab setup, reconnaissance, web assessment, SQL Injection evidence, log-based detection, containment, recovery, findings, and recommended security improvements. The assessment remained within the stated local training environment.

Evidence note: This report preserves the supplied screenshots and project documentation rather than inventing additional test results.